Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 201868138

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       3.715790 d
  Epoch (t0)   1766.1184 BTJD
  Duration     0.1200 d
  Depth        5.881 %  (58,810 ppm)
  SNR          237.3

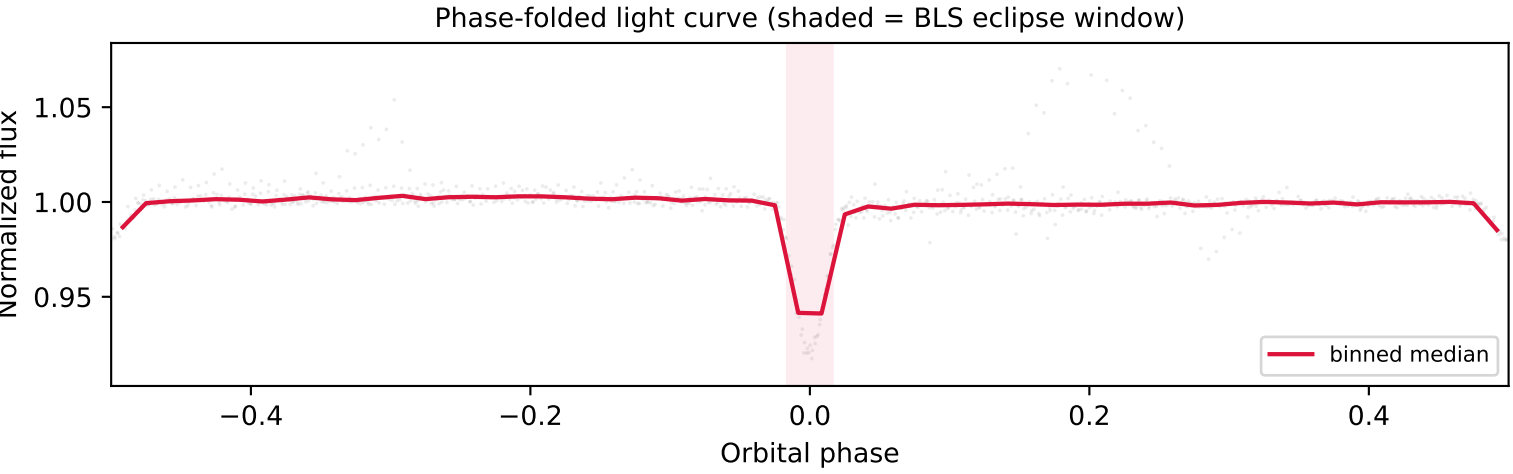
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     6  (gate >= 3)
  Robust depth ratio    1.02  (gate >= 0.50)
  Robust depth sigma    94.6  (gate >= 7.0)
  Odd/even z-score      2.13  (consistent)
  Secondary eclipse     detected, phase 0.49, 5.7 sigma
```

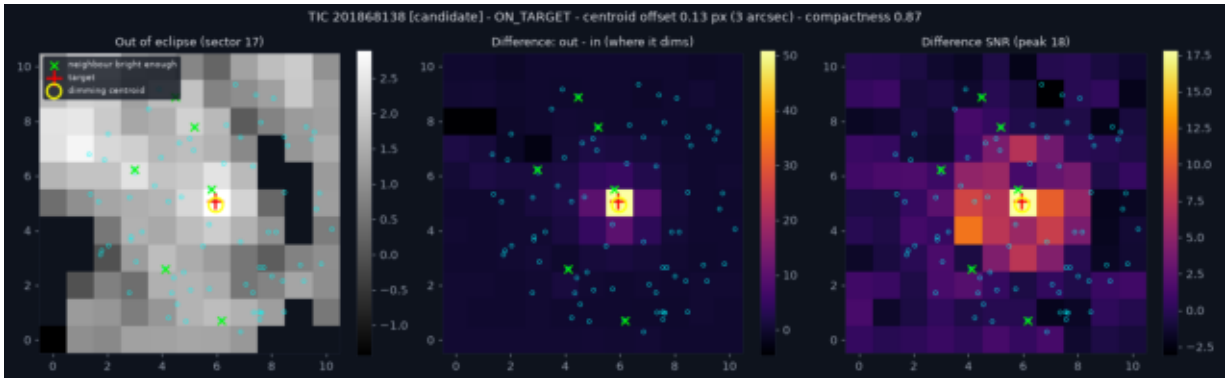
```
CATALOG CROSS-MATCH (novelty)

  VSX                                no match
  ASAS-SN                            no match
  Gaia eclipsing-binary table        no match
  Gaia variable classification       no match
  TESS EB catalog (Prsa+2022)        no match
  ExoFOP TOI list                    no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 17    light-curve dip in this sector: 91.0 sigma    t0 refined by -0.0050 phase
  Difference-image peak SNR   17.83   (gate >= 5.0)      Compactness   0.87   (gate >= 0.50)
  Dimming centroid offset     0.13 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 201868134
    0.59 px away, dTmag +1.95 - farther from the dimming than the target
```





No caveats - clean detection on every check.

TIC 65257021

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period       4.790576 d
  Epoch (t0)   1492.8520 BTJD
  Duration     0.0840 d
  Depth        4.348 %  (43,484 ppm)
  SNR          118.7

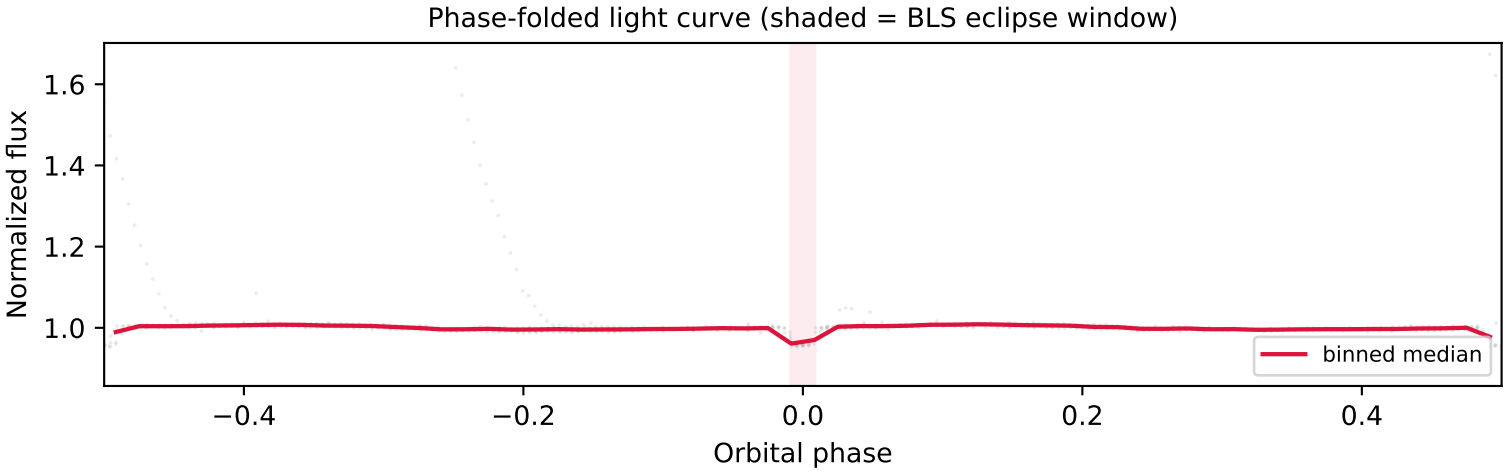
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     4  (gate >= 3)
  Robust depth ratio    0.96  (gate >= 0.50)
  Robust depth sigma    23.1  (gate >= 7.0)
  Odd/even z-score      0.05  (consistent)
  Secondary eclipse     detected, phase 0.49, 5.7 sigma
```

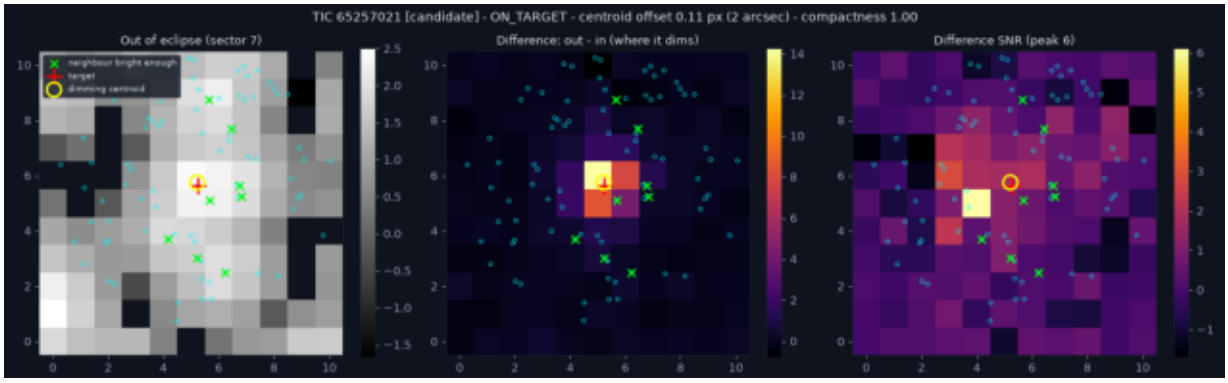
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 20.2 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   6.10   (gate >= 5.0)      Compactness   1.00   (gate >= 0.50)
  Dimming centroid offset     0.11 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 65257017
    0.81 px away, dTmag +1.34 - farther from the dimming than the target
```





No caveats - clean detection on every check.

TIC 468210255

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period        6.706415 d
  Epoch (t0)    1575.3682 BTJD
  Duration      0.1700 d
  Depth         6.926 %  (69,255 ppm)
  SNR           110.0

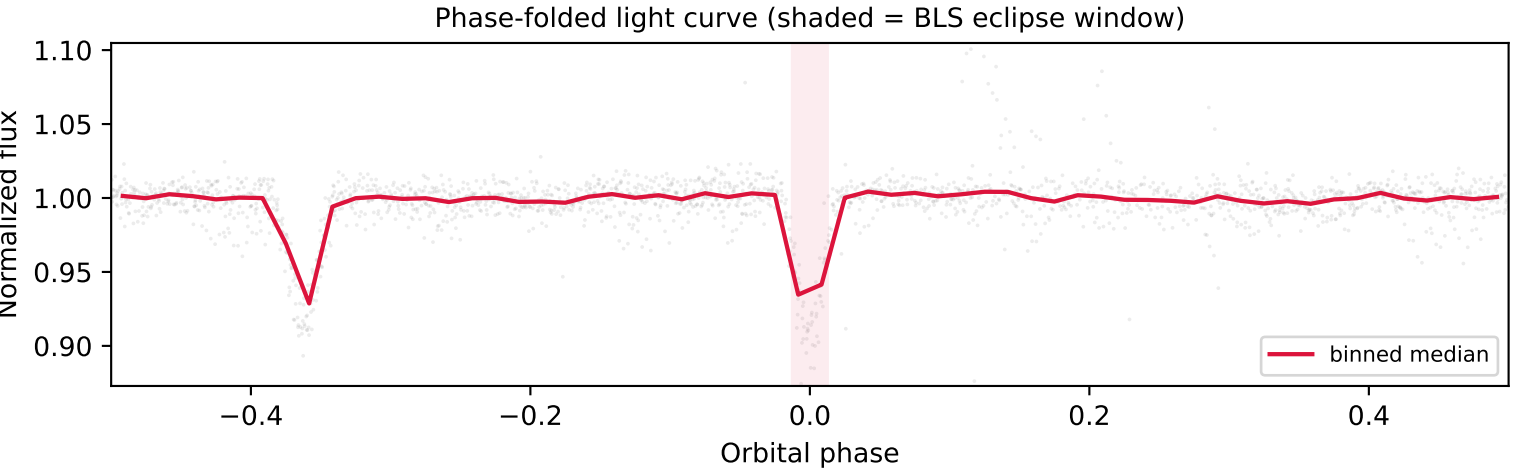
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     7  (gate >= 3)
  Robust depth ratio    1.05  (gate >= 0.50)
  Robust depth sigma    49.9  (gate >= 7.0)
  Odd/even z-score      0.33  (consistent)
  Secondary eclipse     detected, phase -0.36, 7.4 sigma
```

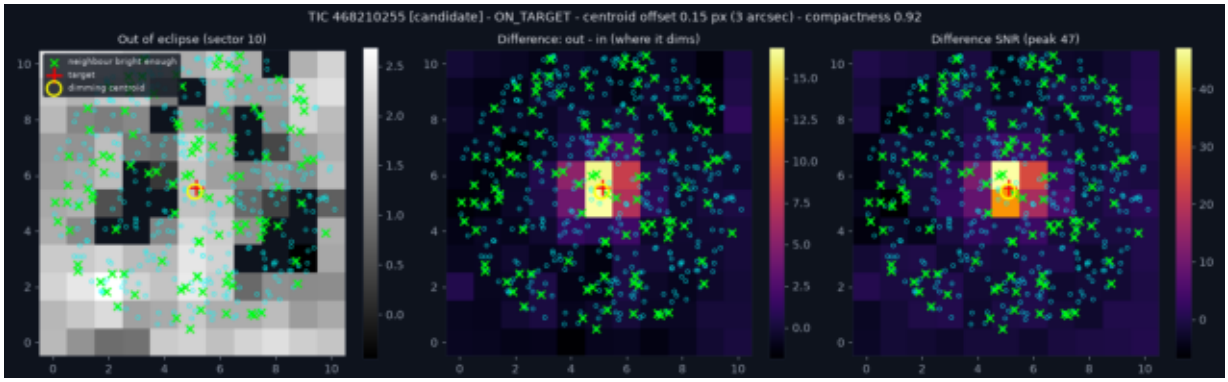
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 37.6 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   47.23   (gate >= 5.0)      Compactness   0.92   (gate >= 0.50)
  Dimming centroid offset     0.15 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 468210271
    0.21 px away, dTmag +1.67 - farther from the dimming than the target
```





Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.064 px (dimming centroid 0.149 px from target vs 0.213 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.